Research activity

Bounding an integral with rectangles — 35 min.

We want to bound $I = \displaystyle\int_0^1 x^2 \, \mathrm{d}x$. For an integer $n \geq 1$, we split $[0, 1]$ into n intervals of equal length. Let L_n be the sum of the areas of the rectangles whose heights are the values of x^2 at the left endpoints, and U_n the sum obtained at the right endpoints.

1. For $n = 4$, draw the two families of rectangles, then calculate L_4 and U_4.
2. Justify that $L_4 \leq I \leq U_4$.
3. Write a Python function with argument n that returns L_n and U_n. Find $U_n - L_n$ algebraically.
4. Find the smallest integer n for which this bound has a width of at most 0.01.
5. Calculate the exact value of I using an antiderivative and check that it lies within the bound obtained for $n = 4$.